The ADER approach for the one-dimensional Euler equations

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Abstract

This paper implements third to fifth-order ADER schemes (ADER3/4/5) for the one-dimensional compressible Euler equations within a finite volume framework. High order accuracy in both space and time is achieved through WENO reconstruction, the Cauchy-Kowalewski procedure, and TT-GRP, all within a single time step and without multi-stage Runge-Kutta updates. On smooth periodic problems, all schemes converge at their design orders across all error norms. On Sod and Lax shock tube problems, shocks, rarefaction waves, and contact discontinuities are captured without significant non-physical oscillations. On the modified shock-turbulence interaction problem, ADER schemes show less numerical dissipation and better resolution of post-shock fine structures than the second-order MUSCL-Hancock scheme at the same grid resolution. Efficiency analysis confirms the advantage of high-order ADER methods in balancing accuracy and cost over long-time simulations.

1. Introduction

In compressible flows, discontinuities such as shock waves may spontaneously form in finite time, even from smooth initial data. This leads to the coexistence of discontinuous structures and smooth structures such as sound waves and density waves, which pose conflicting requirements on numerical schemes: schemes that capture discontinuities satisfactorily tend to over-smooth smooth structures, and vice versa. Godunov's theorem asserts that linear and monotonic schemes cannot exceed first-order accuracy, and so nonlinear schemes are necessary for higher accuracy.

ENO and WENO schemes achieve high accuracy and remain oscillation-free[1][2]. However, for time-discretization, multi-step Runge-Kutta schemes are generally adopted, and for each step, a full reconstruction is required, which increases with accuracy order. ADER, first proposed by Toro et al[3] and later formalized by Titarev and Toro[4], provides a framework for achieving arbitrary accuracy in both space and time in a single time step. ADER comprises three main components: WENO spatial reconstruction for calculating high-order spatial derivatives[5], the Cauchy-Kowalewski procedure for converting spatial derivatives into time derivatives, and the TT-GRP solver[6] for calculating time-averaged fluxes.

In this paper, ADER3/4/5 is implemented using the TT-GRP framework. Convergence is validated on a smooth periodic problem, robustness on the Sod and Lax shock tube problems, performance on mixed smooth-discontinuous flows on the shock-turbulence interaction problem, and finally, an efficiency analysis is conducted on the schemes against MUSCL-Hancock.

2. Methodology

2.1. Background: Governing Equations and Baseline Methods

The next section develops the formulation from the governing equations to the discrete finite-volume update, including the Euler equations, the finite-volume framework, Godunov's first-order scheme, and the second-order MUSCL-Hancock method.

One-dimensional compressible Euler equations

The one-dimensional Euler equations describe the motion of an inviscid compressible gas, expressing conservation of mass, momentum, and energy[7].

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}(\mathbf{U}) = 0 \tag{1}$$

The conserved variable vector U and the flux vector F are respectively:

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ E \end{pmatrix} \qquad \mathbf{F} = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{pmatrix} \tag{2}$$

To close the system of equations, we refer to the ideal gas law, where $\gamma = 1.4$ is the ratio of specific heats.

$$p = (\gamma - 1)\left(E - \frac{1}{2}\rho u^2\right) \tag{3}$$

The eigenvalues of the flux Jacobian $\partial F/\partial U$ are $u-a$, u, and $u+a$, where $a = \sqrt{\gamma p/\rho}$ [7] denotes the local speed of sound. These eigenvalues determine the maximum wave speed and are used in the evaluation of the CFL condition.

Finite-Volume Framework

For this assignment, we will regard then finite volume method as the base method, which partitions the domain into control volumes and integrates the conservation law over each control volume, as opposed to discretizing pointwise[7].

The interval $[x_0, x_1]$ is uniformly partitioned into N cells. The ith cell is defined as:

$$I_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}] \tag{4}$$

Where the cell width is:

$$\Delta x = x_{i+\frac{1}{2}} - x_{i-\frac{1}{2}} \tag{5}$$

And the cell center is:

$$x_i = \frac{1}{2}(x_{i-\frac{1}{2}} + x_{i+\frac{1}{2}}) \tag{6}$$

In the finite volume method, the unknowns are element averages, not point values:

$$\mathbf{U}_i^n = \frac{1}{\Delta x}\int_{I_i} \mathbf{U}(x, t^n)\, dx \tag{7}$$

$\bar{F}_{i+\frac{1}{2}}$ is the time-averaged numerical flux at the cell interface $x_{i+\frac{1}{2}}$.

$$\bar{F}_{i+\frac{1}{2}} = \frac{1}{\Delta t}\int_{t^n}^{t^{n+1}} F(\mathbf{U}(x_{i+\frac{1}{2}}, t))\, dt \tag{8}$$

By integrating Equation (1) over the control volume $I_i \times [t^n, t^{n+1}]$ and applying the divergence theorem, the update formula is obtained directly:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}\left(\bar{F}_{i+\frac{1}{2}} - \bar{F}_{i-\frac{1}{2}}\right) \tag{9}$$

To ensure numerical stability, the time step Δt is restricted by the CFL condition[7], in which $s_{\max}$ is the maximum wave speed and CFL < 1.

$$\Delta t \leq \text{CFL} \cdot \frac{\Delta x}{s_{\max}} \tag{10}$$

The Riemann Problem:

The Riemann problem is a form of initial value problem, consisting of a conservation law and a piecewise-constant initial condition with a single discontinuity located at $x = x_0$[7]. It can be written as:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} = 0 \quad \mathbf{U}(x,0) = \begin{cases} \mathbf{U}_L, & x < x_0 \\ \mathbf{U}_R, & x > x_0 \end{cases} \tag{11}$$

The Riemann Solver:

Solutions to the Riemann problem can be obtained either approximately or exactly. Approximate solvers, such as HLL, HLLC, and Lax-Friedrichs, are computationally efficient but introduce additional numerical dissipation; details can be found in [7]. By contrast, the exact solver resolves all wave structures by iteratively solving for the star-region pressure p^*, described in [7].

$$f(p^*, U_L, U_R) = f_L(p^*, U_L) + f_R(p^*, U_R) + (u_R - u_L) = 0 \tag{12}$$

where for each side $K \in \{L, R\}$:

$$f_K(p^*, u_K) = \begin{cases} (p^* - p_K)\sqrt{\dfrac{A_K}{p^* + B_K}}, & p^* > p_K \text{ (shock)} \\ \dfrac{2c_{s,K}}{\gamma - 1}\left[\left(\dfrac{p^*}{p_K}\right)^{\frac{\gamma-1}{2\gamma}} - 1\right], & p^* \leq p_K \text{ (rarefaction)} \end{cases} \tag{13}$$

with $A_K = \dfrac{2}{(\gamma+1)\rho_K}$ and $B_K = \dfrac{\gamma - 1}{\gamma + 1}p_K$. The root is found via Newton-Raphson iteration [7]:

$$p^*_{(m)} = p^*_{(m-1)} - \frac{f(p^*_{(m-1)}, U_L, U_R)}{f'(p^*_{(m-1)}, U_L, U_R)} \tag{14}$$

As noted by Toro[7], the leading term requires the availability of a classical Riemann solver, exact or approximate. In this work, the exact solver is used as the primary choice, with HLL, HLLC and Lax-Friedrichs employed as a fallback near strong discontinuities.

Godunov's method

The formula (9) for updating needs evaluation of the time-averaged flux $\bar{F}_{i+\frac{1}{2}}$. The issue is how this is done. Godunov's approach is the simplest way of doing this. It is based on assuming piecewise constant data for the solution.

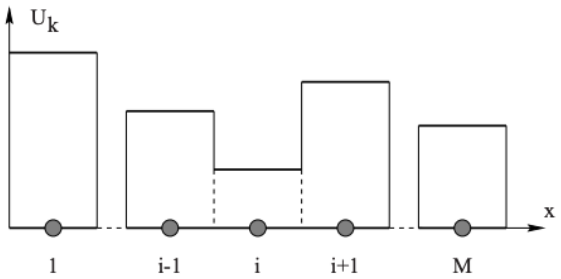

Figure 1: The Godunov's method[7]

At time t^n, Godunov approximates the solution as piecewise constant as illustrated in Figure 1:

$$\mathbf{U}(x, t^n) \approx \mathbf{U}_i^n, \quad x \in I_i. \tag{15}$$

Therefore, at each cell interface $x_{i+1/2}$, there are left and right states $\mathbf{U}_L = \mathbf{U}_i^n$, $\mathbf{U}_R = \mathbf{U}_{i+1}^n$, forming a Riemann problem. Then compute the Riemann problem solution between each neighbouring U_i^n and U_{i+1}^n.

Solving the Riemann problem yields the interface state, and thus the numerical flux is defined as:

$$\mathbf{F}_{i+1/2} = \mathbf{F}\left(\mathbf{U}_{i+1/2}(t^n + 0)\right) \tag{16}$$

Use flux differences to update the conserved quantities:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+1/2} - \mathbf{F}_{i-1/2}\right) \tag{17}$$

Although the classical Godunov scheme is conservative and captures discontinuity, it has two main shortcomings. Because of the piecewise constant approximation in a cell for the solution, it is only first-order accurate in smooth regions, and the error converges only at a rate of $O(\Delta x)$ for refined meshes. In addition, it has a high numerical dissipation error due to the first-order accuracy in the scheme, causing over-smoothing of smooth waveforms and small-scale features, and consequently large error and phase errors in the long-term simulation.

Total Variation Diminishing (TVD) Schemes

A numerical scheme of the form:

$$u_i^{n+1} = H\left(u_{i-k_L}^n, \cdots, u_{i+k_R}^n\right) \tag{18}$$

Is said to be Total Variation Diminishing (TVD) if[7]:

$$TV(u^{n+1}) \leq TV(u^n), \qquad \forall n, \tag{19}$$

Where:

$$TV(u^n) = \sum_i \left|u_{i+1}^n - u_i^n\right|. \tag{20}$$

This ensures that no new local extrema are introduced during the time evolution. Moreover, Godunov's theorem states that the accuracy of a linear monotone scheme is at most first order[7]. In the TVD approach, non-linear limiters are added to the schemes to attain second-order accuracy. However, this is the theoretical limit of the TVD approach.

The MUSCL-Hancock method:

The MUSCL-Hancock scheme is a version of a Monotone Upstream-centred Scheme for Conservation Laws (MUSCL) attributed to Hancock. It is a second-order TVD finite volume method[7].

It uses Godunov's method as a means to update the data:

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}\left(\mathbf{F}(\mathbf{U}_{i+\frac{1}{2}}) - \mathbf{F}(\mathbf{U}_{i-\frac{1}{2}})\right) \tag{21}$$

Here, $\mathbf{U}_{i+\frac{1}{2}}$ is still the solution to a Riemann problem, but now it uses reconstructed states rather than piecewise-constant states:

$$\mathbf{U}_{i+\frac{1}{2}} = \mathbf{U}_{i+\frac{1}{2}}\left(\bar{\mathbf{U}}_i^{R,n+\frac{1}{2}}, \bar{\mathbf{U}}_{i+1}^{L,n+\frac{1}{2}}\right) \tag{22}$$

It is possible to create a non-linear combination of a non-TVD high-order scheme and a first-order monotone scheme, resulting in a TVD scheme of high resolution.

Although the MUSCL-Hancock scheme is second-order accurate for both space and time, it is still limited to second order, and there is a need for higher-order methods, such as ADER.

2.2. The ADER Method

Extension to high order Godunov's method (ADER method)

ADER (Arbitrary High-Order DERivative) is a high-order finite volume scheme[7]. ADER is based on Godunov's method. The basic framework of Godunov's method is retained, but two aspects of Godunov's method have been elevated to a higher order. They are as follows:

- **Higher spatial order**.

  Piecewise constants have been replaced by non-oscillatory higher-order reconstructions, such as WENO/ENO, to obtain polynomial expansions within a unit.

- **High temporal order**.

  The high temporal order is achieved by using a time-averaged value at the interface within a single time step, instead of using multi-stage time updates as in Godunov's method.

The Generalized Riemann Problem

To achieve high temporal order, ADER requires the computation of a high-order time-averaged flux at the interface, which leads to the need to solve a Generalized Riemann Problem (GRP_m) at each cell interface.

The generalized Riemann problem GRP_m is defined as follows[6]:

$$\begin{aligned} &\text{PDEs: } \partial_t \mathbf{Q} + \partial_x \mathbf{F}(\mathbf{Q}) = \mathbf{S}(\mathbf{Q}),\ x \in \mathbb{R},\ t > 0 \\ &\text{ICs: } \mathbf{Q}(x,0) = \begin{cases} \mathbf{P}_i(x), & x < 0, \\ \mathbf{P}_{i+1}(x), & x > 0. \end{cases} \end{aligned} \tag{23}$$

To explain the initial conditions of the GRP_m, on the left side of the interface $x = 0$, the initial value of the reconstructed polynomial $\mathbf{P}_i(x)$ from unit i is used; on the right side, the initial value of the polynomial $\mathbf{P}_{i+1}(x)$ from unit $i + 1$ is used.

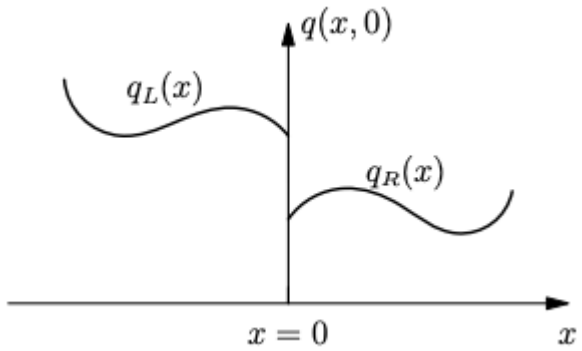

Figure 2: Initial condition for GRP_m[7]

For example, in Figure 2, we have shown the initial condition for GRP_m. Note that in this case, $q_L(x)$ and $q_R(x)$ are polynomials of degree m.

Unlike the classical Riemann problem, in which the initial data are piecewise constants, in the Generalized Riemann Problem, we use polynomials of degree m on both sides of the interface and denote it as GRP_m.

Non-linear spatial reconstruction

In order to reach higher spatial order and recalling Godunov's theorem which requires a non-linear scheme, we propose the use of non-linear spatial reconstruction schemes such as ENO and WENO, which are capable of reaching higher order accuracy without spurious oscillations[7]. These schemes are employed for the computation of the reconstruction polynomials $\mathbf{P}_i(x)$ by the GRP_m at each interface.

At each time step, a spatial reconstruction step is first performed, and by using schemes such as ENO and WENO, an m-degree polynomial $P_i(x)$ is constructed in each cell.

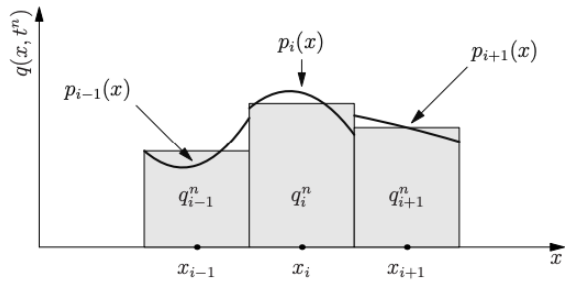

Figure 3: Non-linear spatial reconstruction[7]

As shown in Figure 3, what should first be noted is that $p_i(x)$ in the figure is equivalent to $P_i(x)$ in the above description. At each time step, a polynomial of degree m is constructed in each cell I_i by using non-linear spatial reconstruction schemes such as ENO and WENO.

In our code, we use the WENO-DK scheme[8], which is the classical WENO reconstruction scheme. For an ADER scheme of order M, WENO of order $2M - 1$ is employed: ADER3 adopts WENO5, ADER4 adopts WENO7, and ADER5 adopts WENO9, which is capable of providing spatial derivatives of sufficient accuracy for the subsequent CK procedure. In addition, a positivity-preserving limiter by Zhang-Shu[9] is applied after reconstruction.

The Cauchy–Kowalewski (CK) procedure

The Cauchy-Kovalevski (CK) method transforms the spatial derivatives obtained through WENO into time derivatives using repeated application of the PDEs governing the problem. Following Toro and Titarev[6], for problems involving source terms, the time derivatives take the general form:

$$\partial_t^l \mathbf{Q} = T^{(l)}\Big(\partial_x^0 \mathbf{Q}, \partial_x^1 \mathbf{Q}, \ldots, \partial_x^l \mathbf{Q}, \mathbf{S}^{(l-1)}(\mathbf{Q})\Big) \tag{24}$$

Where $T^{(l)}$ is the nonlinear operator for the l-th order time derivative expressed as spatial derivatives and

lower-order source term contributions through repeated differentiation of the PDEs governing the problem. For the Euler equations used here, $\mathbf{S}(\mathbf{Q}) = 0$, so source term contributions are zero and the expression reduces to:

$$\partial_t^l \mathbf{Q}(x,t) = T^{(l)}\Big(\partial_x^0 \mathbf{Q}, \partial_x^1 \mathbf{Q}, \ldots, \partial_x^l \mathbf{Q}\Big) \tag{25}$$

The CK method for time differencing has been used for orders 2, 3, and 4 for the ADER3, ADER4, and ADER5 methods respectively. For nonlinear problems such as the Euler equations used here, these expressions are obtained through symbolic algebraic manipulations, which become increasingly complicated for higher orders and make the CK method difficult for extension to complex equation sets.

Toro-Titarev solver (TT) for the GRP_m

The spatial derivative is given by WENO, which is then mapped to a time derivative using CK. Finally, the time-averaged flux is obtained using these time derivatives. This is the Toro-Titarev solver for the GRP_m process.

The TT-GRP solver, proposed by Toro and Titarev[6], extends the asymptotic expansion of LeFloch and Raviart [LR], which states that the solution at the interface can be given as:

$$\mathbf{Q}_{LR}(\tau) = \mathbf{Q}(0,0_+) + \sum_{k=1}^{m} \left[\partial_t^{(k)} \mathbf{Q}(0,0_+)\right] \frac{\tau^k}{k!} \tag{26}$$

The leading term in the asymptotic expansion is given by $Q(0,0_+) = G^{(0)}(0)$ and can be obtained by solving a homogeneous Riemann problem. The initial conditions are piecewise constant given by the boundary extrapolation values of the reconstruction polynomials on both sides. The Godunov state $G^{(0)}(0)$ can be found after appropriate sampling of the solution. Any state Riemann solver can be used, in this case, we use the exact solver.

For the higher-order terms in the asymptotic expansion, Goetz and Iske[10] proposed that the Jacobian matrix can be linearised at the Godunov state:

$$\mathbf{A}(\mathbf{Q}) \approx \mathbf{A}(\mathbf{G}^{(0)}(0)) = \mathbf{A}_{LR}^{(0)} \qquad \text{and} \qquad \mathbf{K}^{(l)} = 0 \tag{27}$$

Which leads to a linearised Riemann problem that can be solved to obtain $G^{(l)}(0)$ for any spatial derivative of order l. The Cauchy-Kovalevskaya (CK) procedure can then be used to obtain all Godunov states $G^{(0)}(0), \ldots, G^{(l)}(0)$ in terms of time derivatives using $T^{(l)}$.

$$\mathbf{Q}_{LR}(\tau) = \mathbf{G}^{(0)}(0) + \textstyle\sum_{l=1}^{m} \left[\mathbf{T}^{(l)}\left(\mathbf{G}^{(0)}(0), \ldots, \mathbf{G}^{(l)}(0)\right)\right] \frac{\tau^l}{l!} \tag{28}$$

Substituting $\mathbf{Q}_{LR}(\tau)$ into the flux integral gives the time-averaged numerical flux:

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{\Delta t} \int_0^{\Delta t} \mathbf{F}\left(\mathbf{J}_{i+\frac{1}{2}}(\tau)\right) d\tau \tag{29}$$

For higher-order methods ($m \geq 2$), this integral is evaluated using Gauss-Legendre quadrature: ADER3 uses 1-point, ADER4 uses 2-point, and ADER5 uses 3-point Gauss integration.

3. Numerical Experiments

To measure the performance of the implemented ADER schemes, four numerical experiments are designed: an empirical convergence test, a set of Riemann problem tests, a modified shock-turbulence interaction test, and an efficiency test.

3.1. Empirical convergence test

To measure the convergence of the implemented schemes, a smooth test problem is designed following the suggestion in Toro and Titarev[11]. The one-dimensional Euler equations are solved on the domain $[-1, 1]$ with the following initial conditions:

$$u = p = 1, \rho = 2 + \sin^4(\pi x) \tag{30}$$

And holds periodic boundary conditions and output time $T_{\text{out}} = 2$. Also, the exact solution is known as:

$$u = p = 1, \rho = 2 + \sin^4(\pi(x - t)) \tag{31}$$

Allowing the density error to be measured in the L_1, L_2, and L_∞ norms, respectivey defined as follows:

$$\begin{aligned} L_1 &= \Delta x \sum_{i=1}^{N} \left|u_i - u_i^{\text{exact}}\right| \\ L_2 &= \left(\Delta x \sum_{i=1}^{N} \left|u_i - u_i^{\text{exact}}\right|^2\right)^{1/2} \\ L_\infty &= \max_i \left|u_i - u_i^{\text{exact}}\right| \end{aligned} \tag{32}$$

The convergence rate between two successive resolutions N and $2N$ is computed as:

$$\text{order} = \log_2\left(\frac{E(N)}{E(2N)}\right) \tag{33}$$

A quadrature formula is employed to initialize the cell-averaged data and to compute the exact solution by a cell-averaged integral.

Similar to the original comparison by Titarev and Toro[11], the WENO5-RK3 method is also included as a reference method. It uses a fifth-order WENO spatial reconstruction method and a third-order Runge-Kutta time integration method. Therefore, it is of third-order accuracy. In the present work, four schemes are implemented: ADER3, ADER4, ADER5. They are third-order, fourth-order, and fifth-order ADER schemes using WENO5, WENO7, WENO9 spatial reconstruction schemes, respectively. All schemes are applied to mesh sizes $N = 20, 40, 80, 160, 320, 640, 1280, 2560$.

Table 1: Empirical convergence test for euler equation: IC: $\rho = 2 + \sin^4(\pi x)$. Output Time $T_{out} = 2$, CFL= 0.95, N is the number of cells

Method	N	L_∞ error	L_∞ order	L_1 error	L_1 order	L_2 error	L_2 order
WENO5-RK3	20	2.1210×10^{-1}		1.8659×10^{-1}		1.6491×10^{-1}	
	40	6.7411×10^{-2}	1.65	6.6799×10^{-2}	1.48	5.3844×10^{-2}	1.61
	80	1.0416×10^{-2}	2.69	1.0735×10^{-2}	2.64	8.5470×10^{-3}	2.66
	160	1.3469×10^{-3}	2.95	1.3919×10^{-3}	2.95	1.1099×10^{-3}	2.94
	320	1.6929×10^{-4}	2.99	1.7513×10^{-4}	2.99	1.3959×10^{-4}	2.99
	640	2.1183×10^{-5}	3.00	2.1916×10^{-5}	3.00	1.7468×10^{-5}	3.00
	1280	2.6485×10^{-6}	3.00	2.7402×10^{-6}	3.00	2.1840×10^{-6}	3.00
	2560	3.3108×10^{-7}	3.00	3.4254×10^{-7}	3.00	2.7301×10^{-7}	3.00
ADER3	20	1.4296×10^{-1}		1.3641×10^{-1}		1.1991×10^{-1}	
	40	3.6706×10^{-2}	1.96	3.8802×10^{-2}	1.81	3.0774×10^{-2}	1.96
	80	5.3318×10^{-3}	2.78	5.5212×10^{-3}	2.81	4.4246×10^{-3}	2.80
	160	6.8451×10^{-4}	2.96	7.0993×10^{-4}	2.96	5.6552×10^{-4}	2.97
	320	8.5997×10^{-5}	2.99	8.9012×10^{-5}	3.00	7.0950×10^{-5}	2.99
	640	1.0760×10^{-5}	3.00	1.1134×10^{-5}	3.00	8.8739×10^{-6}	3.00
	1280	1.3452×10^{-6}	3.00	1.3919×10^{-6}	3.00	1.1093×10^{-6}	3.00
	2560	1.6816×10^{-7}	3.00	1.7398×10^{-7}	3.00	1.3867×10^{-7}	3.00
ADER4	20	5.2353×10^{-2}		6.7401×10^{-2}		5.2351×10^{-2}	
	40	2.7671×10^{-3}	4.24	3.3484×10^{-3}	4.33	2.6792×10^{-3}	4.29
	80	9.5826×10^{-5}	4.85	1.1453×10^{-4}	4.87	9.0254×10^{-5}	4.89
	160	3.1408×10^{-6}	4.93	3.7696×10^{-6}	4.93	2.9651×10^{-6}	4.93
	320	1.1244×10^{-7}	4.80	1.3407×10^{-7}	4.81	1.0549×10^{-7}	4.81
	640	4.8717×10^{-9}	4.53	5.7680×10^{-9}	4.54	4.5428×10^{-9}	4.54
	1280	2.6104×10^{-10}	4.22	3.0735×10^{-10}	4.23	2.4223×10^{-10}	4.23
	2560	1.5584×10^{-11}	4.07	1.8180×10^{-10}	4.08	1.4322×10^{-11}	4.08
ADER5	20	5.2053×10^{-2}		6.7330×10^{-2}		5.2295×10^{-2}	
	40	2.7127×10^{-3}	4.26	3.3254×10^{-3}	4.34	2.6663×10^{-3}	4.29
	80	9.3569×10^{-5}	4.86	1.1375×10^{-4}	4.87	8.9154×10^{-5}	4.90
	160	2.9888×10^{-6}	4.97	3.5945×10^{-6}	4.98	2.8258×10^{-6}	4.98
	320	9.3898×10^{-8}	4.99	1.1267×10^{-7}	5.00	8.8604×10^{-8}	5.00
	640	2.9379×10^{-9}	5.00	3.5247×10^{-9}	5.00	2.7709×10^{-9}	5.00
	1280	9.1841×10^{-11}	5.00	1.1016×10^{-10}	5.00	8.6603×10^{-11}	5.00
	2560	3.1832×10^{-12}	4.85	3.4535×10^{-12}	5.00	2.7166×10^{-12}	4.99

Table 1 shows the results. It is found that the ADER3 method and the WENO5-RK3 method have third-order accuracy for all three norms from coarse mesh sizes. The ADER5 method has clean fifth-order accuracy from moderate mesh sizes.

Before moving on to ADER4, it is of interest to consider the stencil of the WENO-DK reconstruction framework[8] for the three schemes considered here. In this framework, the size of the stencil is determined by $k = \lceil M/2 \rceil$, and so the total size of the stencil is given by $2k + 1$. In both ADER3 and ADER5, the size of the stencil is equal to the number of degrees of freedom, and so the system is determined. In ADER4, however, the size of the stencil is five while the number of degrees of freedom is only four, and so the system is overdetermined and solved by least squares. This is true of all odd polynomial degrees in this framework[8].

ADER4 requires finer resolutions before achieving its design order of four, and so only approaches its expected convergence rate near $N = 2560$. This is a known pre-asymptotic effect, and so it is of interest that the overdetermined nature of ADER4 may contribute to this effect, as the extra constraint of the five-cell stencil increases its implicit accuracy beyond four, requiring finer resolutions before achieving the expected convergence rate. However, for a given tolerance, all of the schemes achieve the solution in significantly fewer cells, and so they are clearly beneficial for smooth problems.

3.2. Riemann-problem tests

The robustness of the schemes is assessed on two classical Riemann problems with known exact solutions: the Sod[12] and Lax[13] shock tube tests. The initial conditions are given in Table 2. Godunov, MUSCL-Hancock, ADER3, ADER4, and ADER5 are all compared against the exact solution at the output time T_{out}, allowing a direct comparison between first-, second-, and high-order methods.

Table 2: Initial conditions for the Sod and Lax shock tube tests

Test	ρ_L	u_L	p_L	ρ_R	u_R	p_R	x_0	T_{out}
Sod	1.0	0.0	1.0	0.125	0.0	0.1	0.5	0.2
Lax	0.445	0.698	3.528	0.5	0.0	0.571	0.6	0.14

The results are presented in two ways. Firstly, Figure 4 and 5 will give an overview plot comparing all schemes against the exact solution for both the Sod and Lax tests; then, Figure 6, 7 and 8 will show the three primitive variables ρ u and p are examined individually for each test to allow a more detailed comparison.

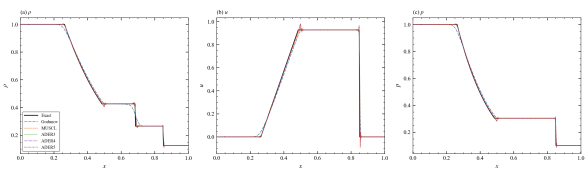

Figure 4: Sod shock tube: ρ, u, p profiles at $T_{\text{out}} = 0.2$, $N = 355$

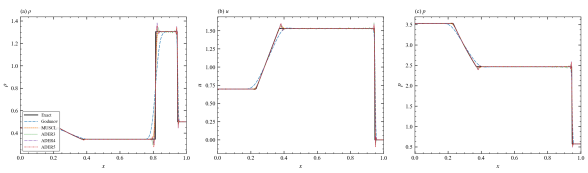

Figure 5: Lax problem: ρ, u, p profiles at $T_{\text{out}} = 0.14$, $N = 355$.

The ρ, u, and p profiles for the Sod and Lax test cases are presented in Figures 4 and 5, respectively. In the Sod test case, all schemes perform well, with ADER3, ADER4, and ADER5 solutions very similar to the exact solution for all three variables. Godunov has the most diffusion, especially at the rarefaction wave and contact discontinuity.

The Lax test case has stronger discontinuities than the Sod test case and thus constitutes a more challenging test. Godunov again has the most diffusion, with significant smearing at the contact discontinuity. ADER3, ADER4, and ADER5 all display small overshoots near the discontinuities, which is a common characteristic of high-order schemes. ADER4 and ADER5 display slightly more overshoot than ADER3, as the WENO stencil of these schemes is less dissipative and thus less effective in reducing oscillations near shocks.

To better understand the performance of the schemes near discontinuities, zoomed-in plots of the ρ, u, and p profiles for the Sod and Lax test cases are presented in Figures 6, 7, and 8.

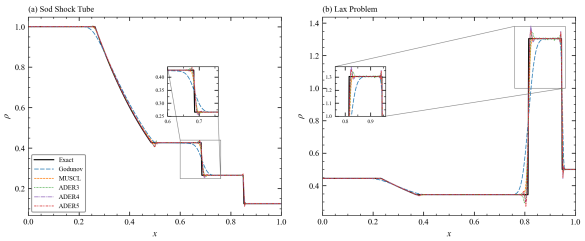

Figure 6: Zoomed-in density ρ profiles for Sod and Lax

The zoomed-in plots of the ρ profiles for the Sod and Lax test cases highlight the performance of the schemes near the contact discontinuity. In the Sod test case, Godunov and MUSCL display significant smearing across the contact discontinuity, while ADER3, ADER4, and ADER5 display very sharp resolution of the discontinuity. In the Lax test case, the stronger shock causes overshoots for ADER4 and ADER5 near the density peak, while ADER3 remains slightly more diffuse.

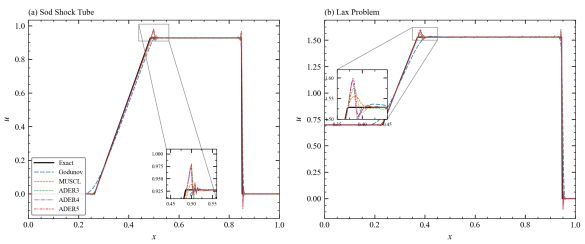

Figure 7: Zoomed-in velocity u profiles for Sod and Lax

The velocity profiles show more clearly the oscillatory behavior near the shock wave. For the Sod problem, all ADER schemes closely follow the exact solution with minimal overshoots. For the Lax problem, ADER4 and ADER5 show large oscillations near the shock wave, whereas ADER3 and MUSCL remain more smooth due to higher numerical diffusion. The Godunov scheme, though having zero oscillations, has large undershoots for the velocity peak, indicating excessive numerical diffusion.

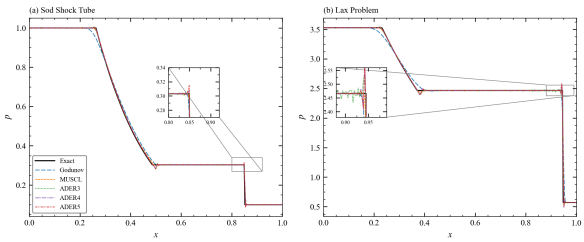

Figure 8: Zoomed-in pressure p profiles for Sod and Lax

The pressure profiles are also well resolved by all schemes for the Sod problem. For the Lax problem, zooming in on the region near the contact discontinuity shows that the Godunov and MUSCL schemes have smeared out the pressure jump significantly, whereas ADER4 and ADER5 resolve it sharply with some oscillations on both sides of the discontinuity. ADER3 strikes a good balance between sharpness and stability.

In both tests, Sod and Lax, it has been clearly demonstrated that higher-order ADER schemes significantly outperform both Godunov and MUSCL schemes in resolving smooth regions and discontinuities. However, some drawbacks remain: a small oscillation near the shock wave, more pronounced for ADER4 and ADER5 due to lower numerical diffusion of the WENO template. ADER3 strikes a good balance between sharpness and stability, whereas the Godunov scheme has maximum diffusion for all profiles.

3.3. Modified shock-turbulence interaction problem

The modified shock-turbulence interaction problem defined by Titarev and Toro[14] is used to validate the schemes' performance in simultaneously handling strong shock waves and smooth structures. The initial conditions are defined by placing a shock at position $x = -4.5$, with post-shock states on the left and a perturbed density field on the right:

$$\begin{bmatrix}\rho_0\\ u_0\\ p_0\end{bmatrix} = \begin{cases}\begin{bmatrix}1.515695\\ 0.523346\\ 1.805000\end{bmatrix}, & x < -4.5,\\ \begin{bmatrix}1+\epsilon\sin(kx)\\ 0.0\\ 1.0\end{bmatrix}, & x > -4.5.\end{cases} \tag{34}$$

Where $\epsilon = 0.1$ and $k = 20\pi$. The computation is run on the domain $[-5, 5]$, discretized by $N = 2000$ cells, and finished at $T_{\text{out}} = 5.0$. The results are plotted on the interval $[-2.5, 3.5]$ and compared with the performance of the MUSCL-Hancock scheme on the same mesh.

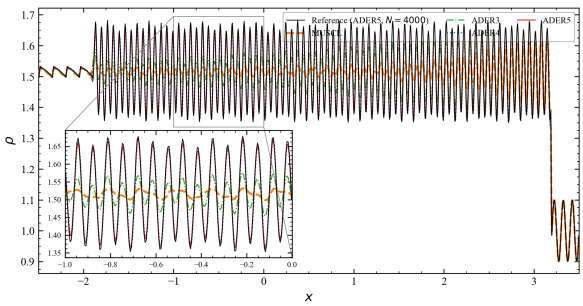

Figure 9: Shock-turbulence interaction, $N = 2000$, $T_{out} = 5.0$

In Figure 9, we plot the density profiles for the modified shock-turbulence interaction problem. Since there is no analytical solution for this problem, a reference solution is generated by running ADER5 on a refined mesh of $N = 4000$ cells, and all other schemes are evaluated relative to this solution. The shock has propagated rightward and interacted with the perturbations, creating complex structures in the post-shock region. The MUSCL-Hancock scheme has under-resolved these structures because of its second-order accuracy and dissipative nature, whereas ADER3, ADER4, and ADER5 on $N = 2000$ cells resolve these features much better.

In the zoomed region of Figure 9, we see how closely each scheme has resolved the post-shock region of the density profile. ADER5 has resolved it best, closely following the reference solution and maintaining both the magnitude and frequency of the oscillations in the post-shock region. ADER4 has also resolved it well, though not as closely as ADER5. ADER3 has slightly more dissipation, though it still resolves much better than MUSCL-Hancock, which has clearly under-resolved these features by damping out much of the oscillatory behavior.

3.4. Efficiency analysis

This section includes two tasks. The first one is to repeat the empirical convergence test from Section 4.1, where the initial conditions will be kept the same, but the final time will be extended to $T_{out} = 1000$, or 500 complete cycles. The comparison of the calculation result of the ADER3, ADER4, ADER5, and MUSCL-Hancock methods on the grid of $N = 100$ cells will be compared against the exact solution. The second task is to analyze the efficiency of these methods by drawing a double logarithmic coordinate, where the error calculated by the L_2 norm is plotted against the time of calculation, and the result will directly reflect the cost of calculation for all methods to obtain the desired accuracy.

The result is presented as follows. Figure 10 shows the result for all methods, where the density profile is calculated using all methods and compared against the exact solution after time $T_{out} = 1000$. Figures 11, 12, and 13 show the efficiency plots for time $T_{out} = 100$, $T_{out} = 200$, and $T_{out} = 400$, respectively, where the error calculated by the L_2 norm is plotted against time calculated on a log-log scale for all methods.

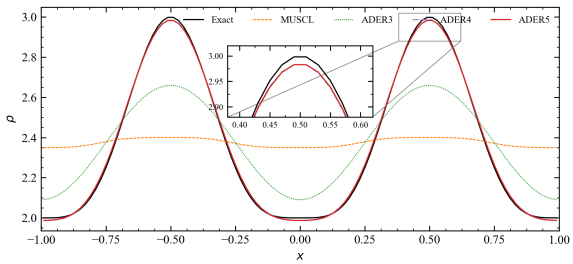

Figure 10: Density ρ profiles at $T_{\text{out}} = 1000$, $N = 100$, CFL = 0.95

Figure 10 shows the result for all methods, where the density profile is calculated using all methods and compared against the exact solution after time $T_{out} = 1000$, where the grid is divided into $N = 100$ cells. The time is equal to 500 complete periods. The most affected is the MUSCL-Hancock method, where almost no waveform is left. The ADER3 method is able to preserve the waveform but loses most of its amplitude. The ADER4 and ADER5 methods preserve the waveform very well, and from the zoomed window, it is obvious that ADER5 is able to preserve the waveform amplitude and phase much better.

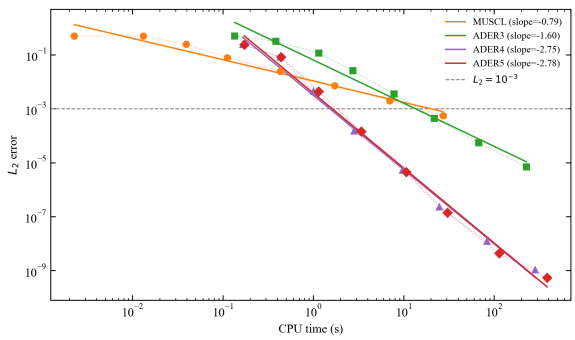

Figure 11: Efficiency plot at $T_{\text{out}} = 100$: L_2 error vs CPU time

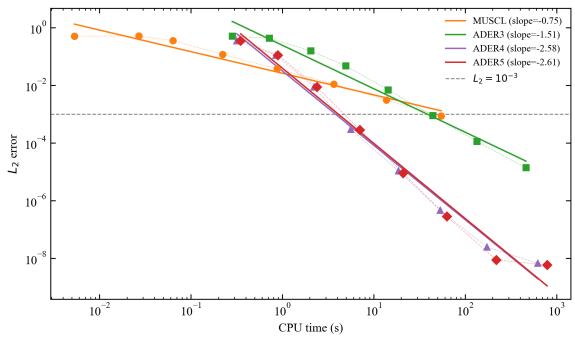

Figure 12: Efficiency plot at $T_{\text{out}} = 200$: L_2 error vs CPU time

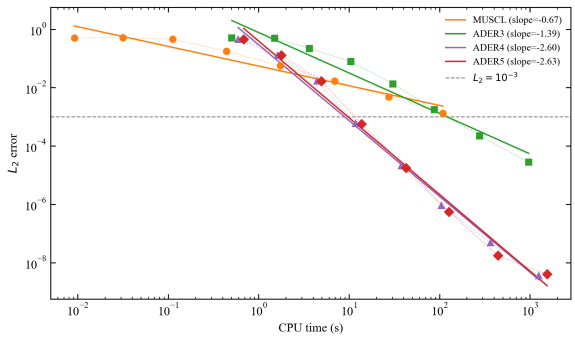

Figure 13: Efficiency plot at $T_{\text{out}} = 400$: L_2 error vs CPU time

Figures 11, 12, and 13 illustrate the efficiency plots for $T_{\text{out}} = 100$, $T_{\text{out}} = 200$, and $T_{\text{out}} = 400$, respectively. The results are clear: all schemes have a steepness in log-log space that increases in magnitude as the order of accuracy increases. Among all schemes, ADER5 has the maximum magnitude of steepness at all T_{out}, followed by ADER4 and then ADER3. The lowest steepness in log-log space is obtained using the MUSCL scheme. This proves that for a given CPU cost, all higher-order accurate schemes have a far greater reduction in L_2 norm accuracy. It should be noted that the accuracy of ADER4 is found to be remarkably close to that of ADER5 in all cases. This can be attributed to the fact that in the case of ADER4, we are using a five-cell stencil in overdetermined mode, which effectively increases its accuracy to fifth order. It should also be noted that as T_{out} increases, CPU times required to attain a desired accuracy of $L_2 = 10^{-3}$ increase significantly in the case of the MUSCL scheme, whereas all ADER schemes are found to be efficient in all cases.

4. Discussion

In this work, four numerical experiments have been carried out to test the performance of ADER schemes. The convergence test on a smooth periodic problem shows that all three ADER methods converge to their expected order of accuracy. One thing worth noting is that ADER4 appears to require more refined meshes before its convergence rate settles. This is, of course, expected, as higher-order methods tend to require more resolution before converging to their expected order of accuracy.

The Sod and Lax tests have shown that ADER methods have sharper resolution of discontinuities compared to Godunov and MUSCL. For near shock cases, ADER4 and ADER5 have small oscillations due to the less dissipative properties of the WENO reconstruction. ADER3, being less dissipative, does not suffer from these oscillations but is still much more accurate than Godunov.

The shock turbulence test is perhaps the most significant test. MUSCL is unable to resolve fine structures after the shock due to the reason that these structures have very small oscillations. ADER4 and ADER5 closely track the reference solution, and one can conclude that for such smooth structures and shock interactions, high-order methods have a big advantage.

The efficiency plots and long time test at $T_{\text{out}} = 1000$ suggest that for smooth problems, MUSCL makes a very poor approximation. The reason is that it takes much more resolution for MUSCL to converge to a similar order of accuracy, and even then, phase and amplitude errors become large. This is why high-order methods are not just more efficient but, for smooth problems, they are the only methods that can be used.

The following remaining part will discuss the potential limitation of the ADER implementation, and also come up with some another alternative choice and discuss for improving the performance of ADER in the

tests we finish in this assignment. In the spatial reconstruction used in this paper, the method used follows the approach proposed by Dumbser et al.[8], which uses three fixed stencils based on a Legendre polynomial basis with arbitrarily chosen linear weights. In odd polynomial cases, e.g., for M = 3 used in the ADER4 scheme, the stencil size exceeds the minimum required for the reconstruction polynomial and becomes an overdetermined problem solved via least squares. This structural difference between even and odd polynomial orders may play a role in the pre-asymptotic behavior found for the ADER4 scheme in the convergence test, where the expected fourth-order convergence rate is approached at finer resolutions than for the ADER3 and ADER5 schemes. Using an alternative reconstruction method based on the standard WENO-JS[5] or WENO-Z[15] formulation, which uses an exactly determined stencil with optimally derived linear weights, may regularize the behavior and allow the ADER4 scheme to enter the asymptotic regime at coarser resolutions.

Another limitation is found within the CK process, and it is associated with its used method of computation. For nonlinear problems, such as the Euler equations, calculating the time derivative using symbolic algebraic computation, as performed by the wxMaxima function, is necessary. The order of the resulting polynomial increases rapidly, and it is difficult to extend such a method for more complex sets of equations. There are several alternative methods that do not require the use of the CK process. The DET solver by Toro et al.[7][16] bypasses the CK process by using local time evolution to compute the polynomial and then using a conventional Riemann solver on Gaussian integration points to compute the inter-cell flux. Another alternative is to replace the CK local evolution by using the Galerkin formula[17]. This method is useful for problems involving stiff source terms, as time evolution is implicit within each cell.

5. Conclusions

This assignment aims to implement and test the ADER schemes for the one-dimensional Euler equations of third, fourth, and fifth order of accuracy. From the numerical experiments conducted in this assignment, the following can be concluded.

For smooth problems, ADER methods attain their theoretical convergence rates and have obvious efficiency advantages compared to second-order methods. For instance, to attain a specific error tolerance, ADER3, ADER4 and ADER5 methods require much less CPU time than MUSCL, and such a difference increases as error tolerance is raised. For long time simulations, the advantages of high-order methods become obvious, as phase and amplitude errors accumulate for low-order methods, eventually rendering them useless.

In the case of discontinuities, the higher-order ADER schemes are better at capturing shock waves and contact discontinuities than the Godunov method and the MUSCL scheme. However, the disadvantage of the higher-order schemes is the presence of slight oscillations around the shock wave. In the ADER4 and ADER5 schemes, the oscillations are more pronounced than in the ADER3 scheme. In the case of strong discontinuities without smooth structures, the above disadvantage may be acceptable.

These are the cases where the high-order schemes are the most important. In the case of the shock turbulence interaction problem, the presence of smooth structures and discontinuities is the main challenge. In the case of long-time simulations, the accumulation of errors is the main concern. The ADER approach is based on the one-step time integration of the TT-GRP method combined with the CK procedure. It is possible to reach high accuracy in both space and time without the need to use the multistage approach of the classical Runge-Kutta method. However, the CK procedure requires the use of symbolic algebraic calculations for the nonlinear systems. It is possible that the applicability of the method is restricted to the Euler equations. CK-free schemes are the next step.

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